

# End-of-Course Projects

## Surgical Scene Understanding

Medical Robotics Course

Academic Year 2025/2026

## 1 Overview

The general topic of these projects is **automatic surgical scene understanding**: the set of computer vision and machine learning techniques that enable a system to “read” what is happening during an operation, typically in laparoscopic and robotic surgery, starting from the endoscopic video stream or from operating-room cameras.

The problem decomposes into a family of complementary sub-tasks, each of which constitutes a project:

- **Workflow analysis** — automatic recognition of the current surgical phase / step / gesture.
- **Anatomy segmentation** — pixel-wise delineation of anatomical structures in the video.
- **Tool detection & tracking** — localisation of surgical instruments and tracking of their trajectory over time.
- **CVS assessment** — automatic evaluation of the *Critical View of Safety*.
- **Body pose estimation** — pose estimation of the operating-room staff (OR-wide), *not* of the instrument articulation.
- **Surgical action recognition** — recognition of fine-grained actions via *action triplets* ⟨instrument, verb, target⟩.
- **Scene reasoning / VQA** — high-level reasoning about the scene via scene graphs and Visual Question Answering.

All projects build on the public datasets of the **CAMMA** research group at the University of Strasbourg (<https://camma.unistra.fr/datasets/>), the international reference for this domain.

## 2 Available datasets

The table below summarises the usable datasets, the tasks they cover, the annotations they provide, and the access conditions.

| Dataset                 | Tasks covered                               | Procedure / Size                                | Annotations                             | Access                        |
|-------------------------|---------------------------------------------|-------------------------------------------------|-----------------------------------------|-------------------------------|
| <b>Cholec80</b>         | Workflow (phase recognition), tool presence | Cholecystectomy, 80 videos, 13 surgeons, 25 fps | Phases (25 fps) + tool presence (1 fps) | Request form; CC-BY-NC-SA 4.0 |
| <b>M2CAI16 workflow</b> | Workflow analysis                           | Cholecystectomy, 41 videos                      | Surgical phases                         | Request form                  |

| Dataset                | Tasks covered                                                                | Procedure / Size                                    | Annotations                                                                | Access         |
|------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------|----------------|
| <b>M2CAI16 tool</b>    | Tool detection                                                               | Cholecystectomy, 15 videos                          | Tool presence                                                              | Request form   |
| <b>MultiBypass140</b>  | Multi-level workflow + adverse-event detection; multi-centric generalisation | Roux-en-Y gastric bypass, 140 videos, multi-centric | Phases, steps, intra-operative adverse events (IAEs)                       | Public GitHub  |
| <b>CholecT50 (T45)</b> | Tool detection + action recognition (triplets)                               | Cholecystectomy, 50 videos                          | 100 triplets ⟨instrument, verb, target⟩ at 1 fps; phases; bbox on 5 videos | Public GitHub  |
| <b>CholecTrack20</b>   | Tool detection & tracking (multi-class multi-tool)                           | Cholecystectomy, 20 videos                          | Multi-perspective trajectories, phases, visual challenges, operators       | Public GitHub  |
| <b>Endoscapes</b>      | Anatomy segmentation, object detection, CVS assessment                       | Cholecystectomy, 201 videos                         | Scene segmentation, detection, CVS assessment                              | Public GitHub  |
| <b>SSG-VQA</b>         | High-level scene understanding (VQA, scene graph)                            | Laparoscopic videos                                 | Geometrically grounded Q&A pairs via scene graphs                          | Public GitHub  |
| <b>MVOR</b>            | Body pose estimation (OR staff, OR-wide)                                     | Real operating room, 3 synchronised RGB-D cameras   | Person bbox, 2D/3D pose, calibration, colour+depth                         | Public GitHub  |
| <b>xawAR16</b>         | Camera tracking / relocalisation (AR in the OR)                              | Operating room, 16 multi-RGBD sequences             | Ground-truth pose of the moving camera                                     | Dedicated page |

### 3 Project descriptions

For each project the task, the reference dataset, a concise description and the learning objectives are given. Groups may agree on variants or extensions with the instructor.

#### 3.1 P1 — Surgical Workflow Analysis

**Dataset:** Cholec80 (baseline) or MultiBypass140 (advanced, optional).

**Description:** develop a model that automatically recognises the current surgical phase from the video stream. The advanced variant on MultiBypass140 extends the problem to hierarchical phase/step recognition and to *cross-center generalisation* (training and testing on different hospi-

tals).

**Learning objectives:** temporal modelling of video sequences; handling of class imbalance; workflow-specific metrics (accuracy, per-phase precision/recall, edit distance); analysis of robustness to domain shift.

### 3.2 P2 — Anatomy Segmentation

**Dataset:** Endoscapes.

**Description:** semantic / instance segmentation of anatomical structures and instruments in endoscopic frames, with the difficulties typical of the domain (reflections, smoke, blood, occlusions).

**Learning objectives:** segmentation architectures; overlap metrics (IoU, Dice); data augmentation specific to surgical imagery; qualitative error analysis.

### 3.3 P3 — Tool Detection & Tracking

**Dataset:** CholecTrack20 (tracking) or CholecT50 / M2CAI16-tool (detection).

**Description:** localise the surgical instruments and follow their trajectory over time, handling entries/exits from the field, occlusions and the simultaneous presence of several instruments.

**Learning objectives:** object detection; temporal association and tracking algorithms (multi-object tracking); MOTA/IDF1 metrics; failure-case analysis.

### 3.4 P4 — Critical View of Safety (CVS) Assessment

**Dataset:** Endoscapes.

**Description:** automatically assess whether and to what extent the three criteria of the *Critical View of Safety* (see lecture by Dr. Mascagni) are satisfied in the video, a task of high clinical relevance for reducing bile-duct injuries.

**Learning objectives:** multi-criterion classification; linking visual perception to clinical safety notions; handling of imbalanced and subjective labels; model interpretability.

### 3.5 P5 — Surgical Action Recognition (Action Triplets)

**Dataset:** CholecT50.

**Description:** recognise fine-grained surgical actions in the form of  $\langle \text{instrument, verb, target} \rangle$  triplets, combining perception of the instrument, the action and the anatomical target.

**Learning objectives:** multi-task and multi-label classification; modelling of relations between entities; average precision metrics per component and per triplet; use of the official splits for benchmarking.

### 3.6 P6 — Operating Room Body Pose Estimation

**Dataset:** MVOR.

**Description:** estimate the 2D/3D pose of the surgical team members from multi-view RGB-D footage of a real operating room, in the presence of strong occlusions and visual clutter. *Note:* here “pose” refers to the staff, not the instruments.

**Learning objectives:** multi-view human pose estimation; use of depth information and camera calibration; pose metrics (PCK, MPJPE); occlusion handling.

### 3.7 P7 — Surgical Scene Reasoning / VQA

**Dataset:** SSG-VQA.

**Description:** build a system that answers questions about the surgical scene (presence, spatial relations, actions) leveraging scene graphs; a project well suited to a multimodal approach.

**Learning objectives:** vision–language integration; structured reasoning over scene graphs; evaluation of VQA systems; analysis of biases in the questions/answers.

## 4 Practical and organisational notes

- **Form-gated data access.** Cholec80 and the two M2CAI16 datasets require a request form to be filled in, and approval is not immediate: groups choosing these datasets must submit the request well in advance of the start of the work.
- **Licence.** Cholec80 is released under the CC-BY-NC-SA 4.0 licence (non-commercial use only): compatible with teaching, but it must be cited correctly.
- **Dataset overlap.** Several datasets share the underlying videos (CholecT50, CholecTrack20, part of Endoscopes and part of M2CAI derive from the Cholec80 pool). Groups combining related datasets must follow the official CAMMA recommendations on handling overlaps ([https://github.com/CAMMA-public/camma\\_dataset\\_overlaps](https://github.com/CAMMA-public/camma_dataset_overlaps)) to avoid train/test contamination.
- **Citations.** Each dataset requires the associated publication to be cited; the references are given on the CAMMA page and in the respective GitHub repositories.
- **Deliverables.** Reproducible code, a short technical report and a final presentation. The metrics to be reported are the standard ones for the task, evaluated on the official splits when available.

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*Dataset reference: CAMMA Research Group, Université de Strasbourg — <https://camma.unistra.fr/datasets/>*